

Lab 2 – Cap Database

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1 Select * Queries

```
1 SELECT *  
2 FROM PEOPLE
```

pid	prefix	firstname	lastname	suffix	homecity	dob
1	Dr. (Hon)	Maynard	Ferguson	Esq.	Montreal	1928-05-04
2	Ms.	Bria	Skonberg		Chilliwack	1987-12-29
3	Mr.	Miles	Davis		Alton	1926-05-26
4	Mr.	Doc	Severinsen		Arlington	1927-07-07
5	Mr.	Louis	Armstrong	Esq.	New Orleans	1901-08-04
6	Ms.	Tine	Helseth		Oslo	1987-08-18
7	Dr.	Cynthia	Robinson	MD	Sacramento	1944-01-12
8	Dr.	James	Morrison	Ph.D.	Oslo	1962-11-11
10	Mr.	Dizzy	Gillespie	III	Montreal	1917-10-21

```
1 SELECT *  
2 FROM Products
```

prodid	name	city	qtyonhand	priceusd
p01	Heisenberg Compensator	Dallas	47	67.76
p02	Universal Translator	Newark	2399	51.50
p03	Apple //+	Duluth	1979	65.02
p04	LCARS module	Duluth	3	17.01
p05	Denis Wick Valve Oil	Dallas	8675309	16.61
p06	PDP-11 operator panel	Beijing	88	88.00
p07	Flux Capacitor	Newark	1007	1.00
p08	HAL 9000 memory chip	Newark	200	1.25
p09	Bach Stradivarius 37	Montreal	1	37900.42

2 Primary key, Candidate key, and Superkey

A super key is a set of fields (at least one) of a table which uniquely identifies each row. A candidate key is a the minimal length of super keys which uniquely identifies each rows. Therefore, every candidate key is a super key but not ever super key is a candidate key. Any key which is of multiple fields is known as a composite key. A primary key is the candidate key which is chosen to unique identify the row. It is also very important to consider the business rules accociated with the database when selecting keys. A primary key should functionally be unique within the associated domain.

3 Data Types in Tables

4 Relational Rules

Making databases ensure relational rules upholds the referential integrity of the data helping to render information. It is good practice to follow these rules as well as infusing business rules into the database itself when possible.

4.1 First Normal Form

The first normal form maintains that there are no multi-valued structure whithin a field or fields relating to a semantically structed need of our business. If a feild has some internal structure that can be pratically separated it should be.

4.2 Access Rows by Content Only

As the databases internal structure is hidden from the user or programmers, accessing data using concepts such as index does not make sense. This makes sense because in postgre, for example, all tables and columns are themselves stored in a database. This means there is no concept of index as the underlying structure likely used is a set.

4.3 Unique Rows

Every row within a table should have a unique primary key rules of which have been previosly mentioned. This is need to differinate and assert no duplication of the same thing within the database.